

Olympiad Test Paper — Science (Class 7)

Chapter 4: Heat

Subject	Science (NCERT Class 7)
Chapter	4 — Heat
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Temperatures are in degrees Celsius (°C) unless a question states otherwise. Thermometer scales are described in words — picture the marks as you read. No calculator is needed — all numbers are designed to work by hand. Most questions need two to four reasoning steps; watch for traps. Do not write on this paper — mark answers on your answer sheet.

1. A small cup of boiling water at 100 °C and a large bucket of warm water at 40 °C are placed side by side. Which statement is correct?

- (A) Both hold exactly the same amount of heat
- (B) The cup has the higher temperature, but the bucket may hold more total heat because it has far more water
- (C) The bucket has the higher temperature but the cup holds more heat
- (D) The cup holds more heat because hotter always means more heat

2. Heat is best described as:

- (A) a measure of how hot or cold a body is
- (B) the average vibration of one atom
- (C) the number marked on a thermometer
- (D) a form of energy that flows from a hotter body to a colder one

3. Two identical mugs are filled — one with 100 mL of water at 90 °C, the other with 200 mL of water at 90 °C. Which has the higher **temperature**?

- (A) the 100 mL mug
- (B) the 200 mL mug
- (C) both are at the same temperature
- (D) the one that cools first

4. When a hot iron rod is dipped into a beaker of cold water and left alone, heat flows:

- (A) from the water to the rod until the rod is hotter
- (B) in both directions equally, so nothing changes

- (C) from the rod to the water until both reach the same temperature
- (D) only if the water is stirred

5. A spoon left in a cup of hot tea slowly becomes warm. This shows that:

- (A) the spoon is making its own heat
- (B) heat flows from the hotter tea to the cooler spoon
- (C) temperature and heat are the same thing
- (D) the tea is gaining heat from the spoon

6. Most thermometers work because the liquid inside them:

- (A) expands (takes up more space) when heated and contracts when cooled
- (B) changes colour when heated
- (C) turns into a gas when heated
- (D) gets heavier when heated

7. When the bulb of a thermometer is warmed, the liquid thread rises up the tube. The thread rises because the liquid:

- (A) expands and needs more room, so it climbs the narrow tube
- (B) is pushed up by air from below
- (C) becomes lighter than the glass
- (D) is boiling inside the bulb

8. The tube of a thermometer is made very **narrow** mainly so that:

- (A) less liquid is needed to fill it
- (B) the liquid cannot escape
- (C) even a small expansion of liquid moves the thread a large, easy-to-read distance
- (D) the glass stays cooler

9. A laboratory thermometer is dipped in melting ice and the liquid settles at the mark halfway between the 0 and 10 lines. The temperature is:

- (A) 5 °C
- (B) 0 °C
- (C) 10 °C
- (D) 15 °C

10. Which range is typical for a **clinical** thermometer used to measure human body temperature?

- (A) 35 °C to 42 °C
- (B) -10 °C to 110 °C
- (C) 0 °C to 100 °C
- (D) 0 °C to 50 °C

11. A clinical thermometer has a sharp narrow bend (a "kink") in its tube. The kink is there to:

- (A) make the thermometer cheaper
- (B) let the liquid expand faster
- (C) stop the liquid thread from slipping back, so the reading holds until you read it
- (D) measure two temperatures at once

12. Why does a **laboratory** thermometer usually have **no kink**?

- (A) Laboratory liquids never expand
- (B) A kink would crack the glass in a lab
- (C) It is read while still in contact with the substance, so the thread must move freely up and down

(D) Laboratory thermometers are always digital

13. A laboratory thermometer is marked from -10°C to 110°C . It would be the wrong tool to measure:

- (A) the temperature of warm milk at 45°C
- (B) the temperature of ice-cold water at 4°C
- (C) the temperature of a beaker of water being heated to 80°C
- (D) the exact body temperature of a feverish patient to one-tenth of a degree

14. Before using a clinical thermometer, you give it a few sharp jerks/shakes. The reason is to:

- (A) warm it up before use
- (B) clean the bulb
- (C) bring the thread back down below normal body temperature so the next reading is correct
- (D) test whether the glass is strong

15. A clinical thermometer reads in the range 35 to 42 with small marks every 0.1 between the numbers. The liquid stops on the 2nd small mark past 37. The reading is:

- (A) 37.0°C
- (B) 39.0°C
- (C) 37.02°C
- (D) 37.2°C

16. The normal temperature of a healthy human body is about:

- (A) 42°C
- (B) 37°C
- (C) 30°C
- (D) 25°C

17. On a laboratory thermometer scale, the big numbered marks are 10 apart and there are 10 equal small divisions between them. Each small division therefore stands for:

- (A) 1°C
- (B) 0.1°C
- (C) 10°C
- (D) 5°C

18. A laboratory thermometer reads three small divisions above the 60 mark, where each small division is 1°C . The temperature is:

- (A) 60°C
- (B) 90°C
- (C) 30°C
- (D) 63°C

19. A patient's clinical thermometer shows the thread halfway between the 38 and 39 marks. The reading is:

- (A) 38.0°C
- (B) 38.5°C
- (C) 39.0°C
- (D) 37.5°C

20. Normal human body temperature of about 37°C is the same as roughly 98.6 on the Fahrenheit scale. A child with a reading of 37°C is therefore:

- (A) at about the normal body temperature
- (B) running a high fever
- (C) dangerously cold
- (D) below the range any thermometer can show

21. Heat travelling through a solid metal rod, from the hot end to the cold end, without the rod itself moving, is an example of:

- (A) convection
- (B) radiation
- (C) conduction
- (D) reflection

22. One end of a metal spoon is held in a flame. After a while the other end becomes hot even though it was never in the flame. This is because:

- (A) the heat radiated across the air gap
- (B) hot metal floated up to the handle
- (C) the spoon created its own heat
- (D) heat was conducted along the metal from the hot end to the cold end

23. Conduction is the **main** way heat travels through:

- (A) liquids only
- (B) solids
- (C) gases only
- (D) empty space

24. Of the four objects below, heat will travel **fastest** by conduction through the:

- (A) wooden ruler
- (B) plastic comb
- (C) rubber band
- (D) iron nail

25. A metal handle and a wooden handle are left in the same warm room for hours. When you touch them, the metal feels **colder** than the wood even though both are at room temperature. This is because:

- (A) metal is actually at a lower temperature than wood
- (B) wood is warmer than the room
- (C) your hand makes mistakes about metal only
- (D) metal conducts heat away from your hand quickly, so your skin cools faster and feels colder

26. A cook stirs hot soup with a metal spoon and leaves it in the pot; soon the spoon's handle is too hot to hold. Replacing it with a wooden spoon helps because wood:

- (A) is a good conductor, spreading heat away from the soup
- (B) is a poor conductor (insulator), so little heat reaches the handle
- (C) absorbs the soup's heat and stores it safely
- (D) reflects the heat back into the soup

27. Heat travelling through water or air by the actual **movement of the heated fluid** is called:

- (A) conduction
- (B) convection
- (C) radiation
- (D) insulation

28. When water in a pan is heated from below, the warm water at the bottom:

- (A) stays at the bottom because it is heavier
- (B) instantly turns to steam
- (C) conducts heat to the top without moving
- (D) becomes lighter, rises, and cooler water sinks to take its place, setting up a circulating current

29. Why is a room heater most effective when placed near the **floor** for heating, while an air-conditioner is fixed **high** on the wall?

- (A) Warm air rises and cool air sinks, so heating from low and cooling from high lets convection currents spread air through the whole room
- (B) It is only about which is cheaper to install
- (C) Electric wires must be near the floor
- (D) The position makes no difference to how the air moves

30. In a beaker of water heated at the centre of the bottom, a few coloured crystals dropped in will trace a path that:

- (A) stays still at the bottom
- (B) shoots straight up and disappears
- (C) rises in the middle and sinks along the sides, showing a convection current
- (D) sinks straight down only

31. Heat from the Sun reaches the Earth across empty space. It travels by:

- (A) conduction, through the air
- (B) convection, through rising hot air
- (C) all three equally
- (D) radiation, which needs no material medium

32. Which mode of heat transfer can take place even through a **vacuum** (no air, no liquid, no solid)?

- (A) conduction
- (B) radiation
- (C) convection
- (D) none — heat always needs a medium

33. Standing near a glowing campfire, you feel warmth on your face even though the hot air rises straight up and does not blow toward you. The warmth you feel reaches you mainly by:

- (A) conduction through the ground
- (B) convection blowing sideways
- (C) the fire pushing cold air away
- (D) radiation travelling straight from the fire to your skin

34. All warm objects give off heat as radiation. Compared with a cool object, a **hotter** object:

- (A) gives off no radiation
- (B) radiates less heat each second
- (C) radiates more heat each second
- (D) radiates the same as a cold object

35. A teacher claims: "Radiation can pass through empty space, but conduction and convection both need some material (matter) to travel through." This statement is:

- (A) false — all three need matter
- (B) false — only convection needs matter
- (C) false — radiation also needs matter
- (D) true

36. A material that lets heat pass through it easily is called a heat **conductor**. Which list contains only good conductors?

- (A) wood, plastic, copper
- (B) air, wool, rubber
- (C) glass, cork, iron
- (D) copper, aluminium, iron

37. A material that does **not** let heat pass through easily is called an **insulator**. Which is the best insulator?

- (A) trapped air (as in wool or a sweater) (B) iron
- (C) aluminium (D) silver

38. Cooking pots are often made with a metal body but a **wooden or plastic handle**. The reason is that the metal body should _____ heat while the handle should _____ heat.

- (A) block; conduct (B) reflect; absorb
- (C) conduct; not conduct (insulate) (D) store; release

39. Which everyday object depends on a material being a **poor** conductor of heat?

- (A) a copper-bottomed cooking pan (B) the metal blade of a knife
- (C) the plastic or wooden handle of a frying pan
- (D) an aluminium cooking foil

40. A jacket "keeps you warm" not by making heat but by:

- (A) being a poor conductor that slows the escape of your body's heat
- (B) producing heat from the cloth (C) reflecting sunlight onto you
- (D) conducting your heat away quickly

41. Two cups of equally hot tea are left out: cup P is bare metal, cup Q is wrapped in a thick woollen cloth. After ten minutes:

- (A) cup Q is hotter, because wool is a poor conductor and slows heat loss
- (B) cup P is hotter (C) both are equally hot
- (D) cup Q is hotter only if the tea was stirred

42. Why do **woollen** clothes keep us warmer than an equal thickness of smooth cotton in winter?

- (A) Wool produces heat from the body's sweat
- (B) Wool traps a lot of air, and trapped air is a poor conductor that slows heat loss
- (C) Wool conducts the body's heat to the surface faster
- (D) Wool is heavier, and heavy things are always warmer

43. Wearing **two thin sweaters** can feel warmer than **one thick one** of the same total thickness because the two-layer arrangement:

- (A) traps an extra layer of air between the sweaters, and trapped air is a poor conductor
- (B) is heavier (C) produces more heat
- (D) reflects more light

44. A bird fluffs up its feathers on a cold day. This keeps it warmer because fluffing:

- (A) traps more air among the feathers, and trapped air slows heat loss
- (B) makes its body produce more heat (C) makes the feathers conduct heat better
- (D) reflects sunlight onto the bird

45. In **summer**, light-coloured (white) clothes are preferred over dark clothes because light colours:

- (A) absorb more heat from sunlight
- (B) make the body produce less heat
- (C) conduct heat to the skin faster
- (D) reflect away most of the sunlight, so the body absorbs less heat and stays cooler

46. In **winter**, dark-coloured clothes can feel warmer in the daytime Sun because dark colours:

- (A) reflect sunlight away
- (B) trap more body heat than light colours
- (C) absorb more of the Sun's heat than light colours do
- (D) produce heat from the Sun's light

47. Two identical metal cans, one painted shiny white and one painted dull black, are left in bright sunshine. After an hour:

- (A) the white can is hotter
- (B) the black can is hotter, because dark surfaces absorb more radiation
- (C) both are at exactly the same temperature
- (D) the white can is hotter only if it is bigger

48. A coastal town has a **sea breeze** during the day. This means that during the day the wind blows:

- (A) from the sea toward the land
- (B) from the land out to the sea
- (C) straight upward only
- (D) there is no wind in the daytime

49. During the **day**, the sea breeze blows from sea to land because:

- (A) the sea heats up faster than the land
- (B) the land heats up faster than the sea, so air rises over the warm land and cooler air flows in from over the sea
- (C) the land stays cooler than the sea all day
- (D) the Sun pulls air toward the water

50. At **night** along the coast, a **land breeze** blows from land to sea. This happens because at night:

- (A) the sea cools faster than the land
- (B) the land cools faster than the sea, so air rises over the warmer sea and cooler air flows out from the land
- (C) the land stays hotter than the sea all night
- (D) there is no temperature difference at night

51. A fisherman notices the wind blowing **out to sea** in the early morning before sunrise. This is a:

- (A) sea breeze, caused by the warm land
- (B) sign of a coming storm only
- (C) breeze that needs the Sun directly overhead
- (D) land breeze, because at night the land has cooled below the sea's temperature

52. Sea breezes and land breezes are both examples of which mode of heat transfer at a large (coastal) scale?

- (A) conduction
- (B) convection
- (C) radiation
- (D) reflection

53. The sea breeze and the land breeze blow in **opposite** directions because:

- (A) the wind always reverses every hour
- (B) by day the land is warmer than the sea, while by night the land is cooler than the sea
- (C) the sea only heats up at night
- (D) the Moon reverses the wind

54. A clinical thermometer is **not** suitable for measuring the temperature of boiling water at 100 °C because:

- (A) its scale only goes up to about 42 °C, far below 100 °C
- (B) its liquid would freeze
- (C) it has no kink
- (D) it is too long

55. To follow the temperature of water as it is heated steadily from cold up to nearly boiling, the right instrument is a:

- (A) clinical thermometer, because of its kink
- (B) clinical thermometer, because it reads in °C
- (C) any thermometer, since all read up to 110 °C
- (D) laboratory thermometer, because its range (about –10 °C to 110 °C) covers the whole rise

56. On a cold morning, the metal frame of a bicycle feels colder to the touch than its rubber grips, although both have been outside all night at the same temperature. The best explanation is:

- (A) the metal is genuinely many degrees colder than the rubber
- (B) the rubber grips were warmed by the Sun overnight
- (C) metal conducts heat away from your hand faster than rubber, so the metal feels colder
- (D) rubber makes its own heat

57. Tea stays hot longer in a thermos flask partly because the flask has a near-**vacuum** between its two walls. The vacuum chiefly stops heat loss by:

- (A) radiation
- (B) reflection of light
- (C) conduction and convection, since both need matter to travel through
- (D) absorption of sunlight

58. A laboratory thermometer reads exactly halfway between the –10 and the 0 marks. The temperature is:

- (A) 5 °C
- (B) –5 °C
- (C) –10 °C
- (D) 0 °C

59. Houses in very hot, sunny regions are often **whitewashed** (painted white) on the outside because a white surface:

- (A) absorbs the most sunlight, warming the house
- (B) conducts heat into the walls quickly
- (C) reflects most of the sunlight, so the house stays cooler inside
- (D) produces a cooling breeze

60. A student lists how the three modes need a medium: "Conduction needs matter; convection needs a moving fluid; radiation needs no medium at all." How many of these three statements are correct?

- (A) all three
- (B) only one
- (C) only two
- (D) none

End of question paper. Maximum marks: 240. Marking: +4 correct, -1 wrong, 0 unattempted.

Solutions — Science (Class 7), Chapter 4: Heat

Marking: +4 correct, -1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	C	21	C	31	D	41	A	51	D
2	D	12	C	22	D	32	B	42	B	52	B
3	C	13	D	23	B	33	D	43	A	53	B
4	C	14	C	24	D	34	C	44	A	54	A
5	B	15	D	25	D	35	D	45	D	55	D
6	A	16	B	26	B	36	D	46	C	56	C
7	A	17	A	27	B	37	A	47	B	57	C
8	C	18	D	28	D	38	C	48	A	58	B
9	A	19	B	29	A	39	C	49	B	59	C
10	A	20	A	30	C	40	A	50	B	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

- (B)** Temperature tells how hot something is; heat is the *total* energy. The cup at 100 °C is hotter (higher temperature), but the bucket holds far more water, so it can hold more total heat even at a lower temperature. "Hotter always means more heat" (D) is the classic trap.
- (D)** Heat is a form of energy that flows from a hotter body to a colder one until both reach the same temperature. The "degree of hotness" (A) defines *temperature*, not heat; (B) and (C) confuse temperature with heat.
- (C)** Temperature is the degree of hotness and does not depend on how much water there is. Both mugs are at 90 °C, so both are at the same temperature — the larger one simply stores more heat.
- (C)** Heat always flows from the hotter object to the colder one. The rod is hotter, so heat flows from rod to water until both reach a common (equal) temperature; it does not flow back to make the rod hotter.
- (B)** The tea is hotter than the spoon, so heat flows from the hotter tea to the cooler spoon, warming the spoon. The spoon makes no heat of its own, and the tea loses (not

gains) heat.

6. (A) A liquid expands (takes up more space) when heated and contracts when cooled. The thermometer turns this volume change into a moving thread; it does not rely on colour change, boiling, or weight change.

7. (A) When warmed, the liquid in the bulb expands and needs more room, so it is forced up the narrow tube, raising the thread. It is not pushed by air or made lighter than glass.

8. (C) A very narrow bore means even a tiny expansion of the liquid pushes the thread a large, easily seen distance, making fine temperatures readable. (Less liquid, no escape, cooler glass are not the main reason.)

9. (A) Halfway between the 0 and 10 marks is 5 °C. (Melting ice is actually 0 °C, but the question fixes the thread's *position* at the midpoint, which reads 5.)

10. (A) A clinical thermometer is built to read human body temperatures, so its scale runs only about 35 °C to 42 °C. The wide -10 to 110 range belongs to a laboratory thermometer.

11. (C) The kink (a sharp narrow bend) stops the liquid thread from slipping back when the thermometer is removed from the body, so the reading stays put until you read it.

12. (C) A laboratory thermometer is read *while still in contact* with the substance, so its thread must rise and fall freely as the temperature changes. A kink would trap the thread, so it is left out.

13. (D) A lab thermometer spans a huge range, so its divisions are coarse (usually 1 °C). It cannot read a fever to one-tenth of a degree the way a clinical thermometer can, so it is the wrong tool for option (D). It handles 4, 45 and 80 °C fine.

14. (C) Shaking jerks the trapped thread back down past the kink, below normal body temperature, so the next patient's reading starts from a low value and is correct. It is not for warming or cleaning.

15. (D) Each small mark is 0.1 °C. Two small marks past 37 means $37 + 2 \times 0.1 = 37.2$ °C. (37.02 wrongly treats each mark as 0.01.)

16. (B) The normal temperature of a healthy human body is about 37 °C (roughly 98.6 °F). 42 °C would be a dangerous fever; 25–30 °C is far too low.

17. (A) Ten equal divisions span 10 °C, so each division = $10 \div 10 = 1$ °C. (0.1 °C would be right only if there were 100 divisions in that gap.)

18. (D) Each small division is 1 °C, so three divisions above 60 means $60 + 3 = 63$ °C. (90 and 30 ignore the stated 1 °C division size.)

- 19. (B)** Halfway between the 38 and 39 marks is $38 + 0.5 = 38.5$ °C.
- 20. (A)** 37 °C is about 98.6 °F, the normal body temperature, so a 37 °C reading means the child is at about normal — not feverish and not dangerously cold.
- 21. (C)** Heat moving through a solid from the hot end to the cold end, while the solid itself stays put, is conduction. Convection needs the material to move; radiation needs no medium.
- 22. (D)** Metal is a good conductor, so heat is conducted along the spoon from the flame end to the cool end, warming the handle. There is no air gap to radiate across along the metal, and the spoon makes no heat.
- 23. (B)** Conduction is the main way heat travels through solids (their particles pass energy along while staying in place). Fluids transfer heat chiefly by convection; empty space, only by radiation.
- 24. (D)** Metals are good conductors and non-metals (wood, plastic, rubber) are poor conductors, so heat travels fastest through the iron nail.
- 25. (D)** Both are at room temperature, but metal conducts heat away from your warm hand quickly, cooling your skin fast, so it **feels** colder. Wood, a poor conductor, draws little heat, so it feels warmer. The metal is not actually colder.
- 26. (B)** Wood is a poor conductor (insulator), so little heat reaches the handle, keeping it cool enough to hold. A good conductor (A) would make the handle hot — the opposite of what helps.
- 27. (B)** Heat carried by the actual movement of a heated fluid (the warm fluid itself rising and circulating) is convection. Conduction passes heat without bulk movement; radiation needs no medium.
- 28. (D)** Heated water at the bottom expands, becomes lighter (less dense), and rises; cooler, denser water sinks to replace it. This sets up a circulating convection current. Warm water does not stay at the bottom.
- 29. (A)** Warm air rises and cool air sinks. Heating from near the floor lets warm air rise and spread; cooling from high on the wall lets cool air sink and circulate. Both use convection currents to mix the whole room.
- 30. (C)** The crystals reveal the convection current: heated water rises in the middle (over the flame) and sinks along the cooler sides, tracing a looping path. It neither stays still nor only sinks.
- 31. (D)** Between the Sun and Earth there is near-empty space with no air or matter to carry heat, so the heat must travel by radiation, which needs no material medium.

Conduction and convection both need matter.

32. (B) Only radiation can cross a vacuum, because it needs no medium. Conduction needs particles in contact and convection needs a moving fluid, so neither works in a vacuum.

33. (D) The hot air rises straight up, so convection is not warming your face; instead, heat radiation travels in straight lines from the fire directly to your skin. That is why you feel warm facing the fire.

34. (C) Every warm object radiates heat, and the hotter it is the more heat it radiates each second. So a hotter object radiates more than a cooler one.

35. (D) The statement is true: radiation can cross empty space, while conduction (through particles) and convection (through a moving fluid) both need matter to travel through.

36. (D) Good conductors are metals: copper, aluminium and iron. The other lists mix in wood, plastic, air, wool, rubber, glass or cork, which are insulators (poor conductors).

37. (A) Trapped air is one of the best everyday insulators (poor conductors), which is why wool and sweaters keep us warm. Iron, aluminium and silver are all good conductors.

38. (C) The body should conduct heat (a metal pot to cook food), while the handle should **not** conduct heat (a wooden/plastic insulator) so it stays cool enough to hold. So: conduct; not conduct (insulate).

39. (C) The plastic or wooden handle works **because** it is a poor conductor, keeping the grip cool. The copper bottom, knife blade and aluminium foil all rely on being good conductors.

40. (A) A jacket makes no heat; it is a poor conductor (with trapped air) that slows the escape of your body's own heat, so you stay warm. It does not produce or reflect heat to warm you.

41. (A) Wool is a poor conductor, so the woollen wrap slows heat loss from cup Q, keeping it hotter than the bare metal cup P after ten minutes. Metal conducts heat away quickly, cooling P faster.

42. (B) Wool's fibres trap a lot of air, and trapped air is a poor conductor, so it slows the escape of body heat — that is why wool feels warmer than smooth cotton. Wool does not make heat or conduct faster.

43. (A) Two thin sweaters trap an extra layer of air between them, and trapped air is a poor conductor, so the layered arrangement slows heat loss better than a single thick layer of the same total thickness.

44. (A) Fluffing feathers traps more air among them, and trapped air is a poor conductor, so it slows the bird's heat loss and keeps it warmer. The bird is not making extra heat or improving conduction.

45. (D) Light (white) clothes reflect away most of the sunlight, so the body absorbs less heat and stays cooler — ideal for summer. Dark clothes would absorb more and feel hotter.

46. (C) Dark colours absorb more of the Sun's heat than light colours, so on a sunny winter day dark clothes feel warmer by soaking up sunlight. They do not reflect it or make heat themselves.

47. (B) Dark, dull surfaces absorb more radiation than shiny white ones, so the black-painted can heats up more in the Sun and ends up hotter than the white can.

48. (A) A sea breeze blows *from the sea toward the land*. (A land breeze would blow from land to sea — the opposite.)

49. (B) By day the land heats up faster than the sea. Air rises over the warm land, and cooler air flows in from over the sea to take its place — that incoming wind is the sea breeze (sea → land).

50. (B) At night the land cools faster than the sea, so now the sea is warmer. Air rises over the warmer sea and cooler air flows out from the land — a land breeze (land → sea).

51. (D) Wind blowing out to sea before sunrise is a land breeze: overnight the land has cooled below the sea's temperature, so air flows from the cooler land toward the warmer sea. It is not a sea breeze and does not need the Sun overhead.

52. (B) Both breezes are caused by warm air rising and cooler air flowing in to replace it, which is convection — here acting at the large coastal scale.

53. (B) The direction depends on which surface is warmer. By day the land is warmer than the sea (sea breeze, sea → land); by night the land is cooler than the sea (land breeze, land → sea). The reversed temperature difference reverses the wind.

54. (A) A clinical thermometer's scale stops at about 42 °C, far below 100 °C, so it simply cannot show boiling-water temperature (and the heat could burst it). The kink is unrelated to this limit.

55. (D) The water rises from cold to nearly boiling, so you need the wide range of a laboratory thermometer (about −10 °C to 110 °C). A clinical thermometer (35–42 °C) cannot cover that range, and not all thermometers read to 110 °C.

56. (C) Both are at the same temperature, but metal conducts heat away from your hand much faster than rubber, cooling your skin quickly, so the metal *feels* colder. The metal is not actually colder, and rubber makes no heat.

57. (C) Conduction and convection both need matter to travel through, so removing the matter (a vacuum between the walls) blocks those two paths and greatly slows heat loss. (Radiation is cut separately by the silvered, shiny walls.)

58. (B) Halfway between the -10 and 0 marks is -5 °C. (Reading it as $+5$ ignores the minus sign — a designed trap.)

59. (C) A white surface reflects most of the sunlight instead of absorbing it, so a whitewashed house absorbs less heat and stays cooler inside in hot, sunny regions.

60. (A) All three statements are correct: conduction needs matter (particles in contact), convection needs a moving fluid, and radiation needs no medium at all. So the count is three.

Solutions complete: 60 / 60.